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RESEARCH ARTICLE

A Recommendation System Based on Fuzzy Signature

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ABSTRACT In recent years, recommendation systems have become essential tools for managing the overwhelming volume of information users face daily. In the context of Technology Enhanced Learning (TEL), for instance, where learners can be faced with a large variety of resources, their ability to deliver accurate suggestions is crucial for enhancing user experience, learning engagement, motivations, and overall learning outcomes. This paper introduces a novel recommendation system based on Fuzzy Signatures. A Fuzzy Signature is a fuzzy relation representing user interests. A similarity metric, named user kindredness, is proposed to determine like-minded individuals who probably share same interests. The kindredness is used to establish user neighborhoods from which the recommendations are generated. For ease of comparison, the performance of the method is evaluated on the Movielens dataset and compared against state-of-the-art approaches, including collaborative filtering, probabilistic methods, and fuzzy genetic algorithms, demonstrating good improvements in terms of accuracy and error rates.

INDEX TERMS Collaborative filtering, content-based filtering, fuzzy user signature, user model.

NOMENCLATURE

FUS	Fuzzy User Signature.	R	R Set of possible ratings in the recommender system.
CF	Collaborative Filtering.	Mat	User–Item–Rating matrix.
PF	Probabilistic Filtering.	Mat_{u_i}	A 2-D User-specific slice of Mat .
GIM	Fuzzy Genetic Method.	$SF(s_j)$	Rating Frequency.
TS	Tagging System.	$SI(m_k)$	Rating per Item.
U	Set of users in TS .	$RatPop_{u_i}(s)$	Rating Popularity.
T	Set of tags in TS .	$ItemAttr_{u_i}(m)$	Item Attractiveness.
R	Set of resources in TS .	$US_{u_i}(s, m)$	User Signature.
Y	Set of relationships ($U \times T \times R$).	$UK(u_i, u_j)$	User Kindredness.
$ResAttr_{u_k}(r)$	Resource Attractiveness.	$p_{u_i}^m$	Predicted rating of user u_i for item m (Resnick formula).
#	“Number”, as in number of elements.	$\bar{r}_{u_i}$	Average rating of user u_i .
$TagPop_{u_k}(t)$	Tag Popularity.	MAE	Mean Absolute Error.
$US_{u_k}(r, t)$	User Signature.	RMSE	Root Mean Square Error.
$N_{u_i}^k$	Top- k neighborhood of user u_i .	MAE_{data}	MAE calculated per dataset.
M	Set of items in the recommender system.	MAE_i	MAE calculated per user.
		MAE_{users}	Average of MAE across all users.

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$RMSE_{users}$	Average of RMSE across all users.
CR	Coverage Rate.
V	Total set of predictions attempted.
$\hat{Y}$	Subset of successful predictions.

I. INTRODUCTION

In the current era of human-AI interactions [1], recommendation systems have become increasingly essential in various sectors, including e-commerce [2], [3], entertainment [4], tourism [5], [6] and e-learning [7]. These systems play a crucial role in addressing the information overload problem. As an example, in e-learning environments, this challenge can be mitigated by guiding learners through the vast array of educational materials—ranging from full courses to individual lessons and learning objects [8], [9]. Another common application is in recommending movies and music, where systems help users discover new content tailored to their tastes and preferences. The success of recommendation systems largely depends on their ability to understand user preferences, a task complicated by the vast and diverse nature of the data. Traditional models often fall short in accurately predicting preferences, particularly in cases where user data is sparse, resulting in less effective recommendations and a poorer user experience [10].

To address these problems, this article presents an innovative method based on Fuzzy User Signatures (FUSs) [11], a fuzzy-based model that captures and represents user preferences. To the best of our knowledge, this is the first time the Fuzzy Signature model has been applied to a recommendation system. The aim of this approach is to enhance the precision and relevance of recommendations by analyzing detailed patterns of user behavior. Unlike traditional collaborative filtering methods, FUS, structured as a matrix, allows for a more sophisticated and statistically robust calculation of user similarities, referred to as “kindredness” [11]. Specifically, the proposed method explicitly models individual rating styles and introduces an asymmetric comparison mechanism, designed to be more robust against sparse data while better capturing the subjective nature of user preferences. Fuzzy logic is particularly well-suited for this task as user preferences and needs are often ambiguous and difficult to quantify in precise terms. The use of fuzzy-based models enables a more nuanced representation of user interests, capturing the inherent uncertainty and gradation of preferences, which are often challenging to express with traditional crisp values.

The system's effectiveness was tested against other recent techniques, including Probabilistic Filtering (PF) [12], Fuzzy-Genetic (GIM) method [13], and the widely-used Collaborative Filtering (CF) [14], on the MovieLens dataset, as it is commonly used as a benchmark for recommender systems. The results demonstrate that employing FUSs offers a good improvement, suggesting this approach as a promising path for future advancements in recommendation technology. Moreover, the proposed method is flexible and can potentially

be applied to different domains of recommender systems, such as e-commerce, entertainment, tourism and e-learning.

The rest of the paper is organized as follows: Section II explains the theoretical background introducing Fuzzy User Signature. Section III provides an overview of the related works in the field of recommendation systems. In Section IV, the proposed FUS method is described. The experimental setup is described in Section V, covering the dataset, evaluation methodology, validation metrics, and results. It also presents a comparative analysis of our approach against PF, GIM, and CF, followed by a discussion of the key findings. Section VI closes the paper with conclusions and future research prospects.

II. THEORETICAL BACKGROUND: FUZZY USER SIGNATURE

Yager and Reformat [11] defined a fuzzy relation representing the activities of a user in a tagging system, named Fuzzy User Signature. The relation represents the activities of a user in terms of tagging actions with respect to a set of resources R available in the tagging system. The main application of fuzzy signature is to find users with similar interests and to find groups of similar users. Formally, a tagging system is a tuple, defined as follows:

$$TS = (U, T, R, Y) \quad (1)$$

where U is the set of users, T is the set of tags, R is the set of resources and Y is the relationship ($U \times T \times R$). In particular, a triple $(u_k, t_i, r_j) \in Y$ means that the user u_k assigned the tag t_i to the resource r_j .

All the user activities in the tagging system can be represented as a 3-D matrix with three dimensions: users, resources, tags. If a user u_k adds a tag t_i to the resource r_j , a point in the 3-D matrix $\langle U, R, T \rangle$ with coordinates $\langle k, j, i \rangle$ is set. A slice of the matrix for the user u_k is a 2-D matrix with the resources on the columns and the tags on the rows. The cells of this matrix will contain a point in position (j, i) if user u_k used the tag t_i for the resource r_j .

Considering the 2-D matrix for a fixed user u_k , it is possible to define the following two fuzzy sets, namely Resource Attractiveness $ResAttr$ and Tag Popularity $TagPop$.

$$ResAttr_{u_k}(r) = \left\{ \frac{b_1}{r_1}, \frac{b_2}{r_2}, \dots, \frac{b_j}{r_j}, \dots, \frac{b_J}{r_J} \right\} \quad (2)$$

where

$$b_j = \frac{\# \text{ of tags used for } r_j \text{ and } u_k}{\max \# \text{ of diff. tags used for a single res. by } u_k} \quad (3)$$

$ResAttr$ represents the degree of attention that a user has with respect to each resource. Its value is given by the ratio between the number of tags used by the user u_k for a certain resource r_j , and the maximum number of tags used to tag a single resource out of all those tagged by the user u_k .

$$TagPop_{u_k}(t) = \left\{ \frac{a_1}{t_1}, \frac{a_2}{t_2}, \dots, \frac{a_i}{t_i}, \dots, \frac{a_I}{t_I} \right\} \quad (4)$$

where

$$a_i = \frac{\# \text{ of times } t_i \text{ is used by } u_k}{\max \# \text{ of res. tagged by } u_k \text{ with a single tag}} \quad (5)$$

TagPop represents the relative frequency of the tag compared to all the others. Its value is given by the ratio between the number of times user u_k has used tag t_i , and the highest number of times u_k has used any single tag. Lastly, it is possible to define a fuzzy relation that captures both aspects (popularity of tags and attractiveness of resources) in a unique value, namely the UserSignature (US) relationship, defined as:

$$US_{u_k}(r, t) = ResAttr_{u_k}(r) \times TagPop_{u_k}(t) \quad (6)$$

and for a specific resource r_j and t_i ,

$$US_{u_k}(r_j, t_i) = \min\{ResAttr_{u_k}(r_i), TagPop_{u_k}(t_j)\} \quad (7)$$

High values of *US* indicate resources that are of interest to the user, labeled with tags that the user likes. In this sense, it represents the interests of the user and can be considered as a concise model of the user interest, i.e., a signature of the user in the tagging system.

III. RELATED WORKS

Recommendation (or recommender) systems rely on information regarding the user's activities to suggest a set of items he/she might be interested in. Since the surge in research interest in recommendation systems in the mid-1990s [15], numerous techniques have been developed to specify the information to be extracted and the methods for generating recommendations. These are based on four main approaches that will be discussed in the following sections — collaborative filtering, content-based filtering, knowledge-based filtering, and hybrid filtering.

A. COLLABORATIVE FILTERING

Recommendation systems based on collaborative filtering focus on the ratings that users assign to the items. Each user is described by a vector containing all the ratings he/she provided, and these vectors are stored in the *user/item evaluation matrix*. In order to generate recommendations, the system predicts the missing ratings based on the ratings provided by other users. This step can be performed applying one of two main categories of algorithms—*memory-based* and *model-based*. The former is based on a measure of similarity following a user-based approach [16], an item-based approach [17], or a hybridization of both [18]. The latter is based on data mining and machine learning algorithms that train predictive models starting from the evaluation matrix; the most used techniques include clustering [19], sparse factor analysis [20], and dimensionality reduction [21].

The main advantage offered by collaborative filtering is not requiring additional data about users or items, whereas the main disadvantages are low performance when providing recommendations to new users with few ratings, i.e., it suffers the cold start problem and, in the case of model-based approaches, the recommendations can be difficult to explain.

B. CONTENT-BASED FILTERING

Content-based filtering recommendation systems focus on additional information about the items to suggest new items that are similar to those the user already interacted with. Using the specific user's ratings and the attributes of the items, a model is trained in order to predict the user's level of appreciation for items he/she has not yet rated or purchased. The main difference with collaborative filtering is that the other users' ratings do not contribute to the suggestions. Some specific applications based on this approach include semantic web mining for an online shopping recommendation system [22], and a recommendation system for relevant information in an online e-learning environment [23].

The main advantages of this approach are the ability to recommend new items for which few to no ratings are provided, and to properly recommend new items. The main disadvantages are the lack of novelty in the items suggested, since they have to be similar to the ones previously purchased, and the inability to provide recommendations to new users.

C. KNOWLEDGE-BASED FILTERING

Knowledge-based filtering recommendation systems focus on information provided by the user as constraints or requirements combined with domain knowledge provided by experts and item descriptions to suggest items that fit the user's needs.

This approach is not based on user ratings, neither on the evaluation history of the user. Moreover, it does not suffer the cold start problem, since user requirements and domain knowledge do not need a large dataset to be obtained.

Two interesting applications of this approach include a recommendation system that analyzes review texts to suggest new restaurants to tourists that are almost identical to those they have enjoyed in the past, based on the feedback received, [24], and a recommendation system based on ontologies and sentiment analysis to detect stressed or depressed users and send happy, calm and motivational messages [25].

D. HYBRID FILTERING

Recommendation systems based on hybrid filtering combine two or more of the approaches described above to achieve better performance and mitigate the disadvantages of individual methods. Over the years, numerous hybridization techniques have been developed, such as

- *weighted* - the two approaches are combined with two weights that are gradually adjusted [26]
- *switching* - the utilized approached is switched according to the situation [27]
- *cascaded* - the model created by the first approach is combined with another one to create a set of candidates with similar tastes [28]
- *mixed* - two recommendation approaches are utilized together to generate a large number of recommendations [29]

- *feature combination/augmentation* - certain features are produced by one recommendation technique and applied to another technique, generating recommendations that employ additional functions [30], [31]

IV. A RECOMMENDATION METHOD BASED ON FUZZY USER SIGNATURE

This work introduces a novel hybrid recommendation method - FUS - based on Fuzzy User Signatures [11]. The FUS method consists of two main phases, depicted in Fig. 1: Offline Batch Processing and On-Line Processing.

In the first phase, information about users, movies and ratings is fetched from the application database in order to calculate for each user u_i the *Fuzzy User Signature* US_{u_i} , a fuzzy relation that describes the activities of the user. Then, for each ordered pair of users (u_i, u_j) , the *Similarity* $sim(u_i, u_j)$ is calculated using US_{u_i} and US_{u_j} . The similarity $sim(u_i, u_j)$ is a value in the range $[0, 1]$ that represents how u_i and u_j are similar in terms of equal ratings assigned to the same movies from the point of view of the user u_i . Finally, a User Model is built as a dictionary consisting of pairs $(u_i, \text{list of similar users})$, i.e., for each user, a set of similar users according to the metric $sim(\cdot)$ is stored. This batch process, described in section IV-A, is performed periodically over a fixed time interval (e.g., once a day), because it represents the most computationally demanding part of our method. Moreover, the information about users, movies and ratings evolves relatively slowly and the results can be efficiently reused.

In the second phase, the recommendations are generated in real time when a user u_i interacts with the application. The list of similarities between u_i and the other users is extracted from the User Model, and the top-k neighbors are retrieved based on the highest similarity values, creating the neighborhood $N_{u_i}^k$ that contains the top-k neighbors. The value k is the number of neighbors and is the main parameter of the FUS method, identified empirically according to the characteristics of the specific application. Its influence on performance will be evaluated in Section V. Finally, the movies recommended to the user u_i are obtained from the neighborhood by calculating, for each movie m , the predicted rating that u_i would assign to m ; the top-x movies with the highest predicted score are extracted. As a result, each user receives an updated and customized set of recommendations. This final phase can be efficiently executed in real time, since the computations involve a small number of users (the neighborhood). This ensures that the system remains highly responsive.

In summary, the approach creates a feedback loop where applications feed data into the application database, which is then processed to generate user-specific recommendations, as shown in Fig. 1.

A. A FUZZY SIGNATURE-BASED USER MODEL

In traditional CF methods, the user model is simply represented by a rating vector with all the ratings the user has

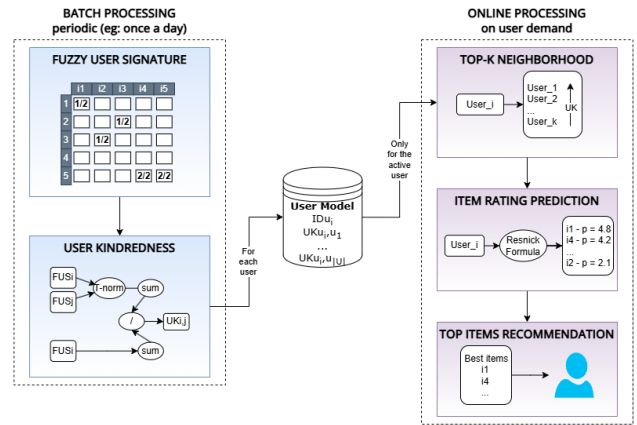


FIGURE 1. Fuzzy user signature-based recommendation approach.

TABLE 1. Matrix representing the rating activity of a user.

	m_1	m_2	m_3	m_4	m_5	m_6	m_7	m_8	m_9	m_{10}	Rating frequency
1	x										1
2			x						x		2
3						x					1
4					x					x	2
5				x			x	x			3
Rating per item	1	0	1	1	1	1	1	1	1	1	

given to items. In the proposed method, we use the Fuzzy User Signature as the user model of the recommendation technique. This model simultaneously considers the ratings used by the user and the items he/she is interested in. In fact, it will be possible to quickly find users with similar interests – as they have given similar ratings to the same items – and also to consider the fact that users may have a different scale for the ratings – i.e., some users prefer to give higher ratings than others.

Let $U = \{u_1, u_2, \dots, u_{|U|}\}$ be the set of all users, let $M = \{m_1, m_2, \dots, m_{|M|}\}$ be the set of all items, and let $S = \{s_1, s_2, \dots, s_{|S|}\}$ be the set of all ratings. Usually, the set S contains a numerical ordinal scale (e.g., 1 to 5). All the ratings given by the users to the items can be represented by a 3-D matrix $\mathbf{Mat} = \langle U \times S \times M \rangle$. A point of the matrix, $\mathbf{Mat}(i, j, k)$ represents that the user $u_i \in U$ gave the rating $s_j \in S$ to the item $m_k \in M$.

A 2-D slice of the matrix, fixing a user u_i , contains all the ratings given by the user u_i to all the items in M , $\mathbf{Mat}_{u_i} = \langle S \times M \rangle$, where $\mathbf{Mat}_{u_i}(j, k)$ represents that the user rated the item m_k with the rating s_j . Table 1 represents the matrix $\mathbf{Mat}_{u_i}$ with the ratings given by the user u_i . In particular, the rows contain the ratings, whereas the columns contain the items.

The counting of marks on a single row s_k represents the number of times a specific rating has been given by the user u_i . This value is defined as the Rating Frequency SF

$$SF(s_j) = \sum_{k=1}^{|M|} (\mathbf{Mat}_{u_i}(j, k)) \quad (8)$$

In the same way, for a single column m_k , it is possible to define the value Rating per Item SI

$$SI(m_k) = \sum_{j=1}^{|S|} (Mat_{u_i}(j, k)) \quad (9)$$

In the FUS method the first step in computing the Fuzzy User Signature from the user matrix Mat_{u_i} consists in applying the two measures SF and SI to derive two fuzzy sets: Rating Popularity $RatPop_{u_i}$ and Item Attractiveness $ItemAttr_{u_i}$.

$RatPop_{u_i}$ represents the degree of utilization or popularity of a rating with respect to all the others for the user u_i . The membership function is defined as follows:

$$RatPop_{u_i}(s) = \left\{ \frac{a_1}{s_1}, \frac{a_2}{s_2}, \dots, \frac{a_j}{s_j}, \dots, \frac{a_{|S|}}{s_{|S|}} \right\} \quad (10)$$

where

$$a_j = \frac{SF(s_j)}{\max_{s \in S} SF(s)} \quad (11)$$

The membership values a_j are calculated in reference to the maximum number of items rated with the rate $s_j \in S$. However, other numbers can be used at the denominator in Eq. 11, as the average number of items rated with s_j , or the median, with a slightly different meaning [11]. As an example, considering the matrix Mat_{u_i} of Table 1, using Eq. 10 the following fuzzy set $RatPop$ is obtained:

$$RatPop_{u_i}(s) = \left\{ \frac{1/3}{1}, \frac{2/3}{2}, \frac{1/3}{3}, \frac{3/3}{4}, \frac{3/3}{5} \right\}$$

The fuzzy set $RatPop_{u_i}$ can be used to understand if the user u_i is inclined to use higher or lower scores to rate the items.

Similarly, $ItemAttr_{u_i}$ represents the degree of interest of the user u_i in each item. It is defined as follows:

$$ItemAttr_{u_i}(m) = \left\{ \frac{b_1}{m_1}, \frac{b_2}{m_2}, \dots, \frac{b_k}{m_k}, \dots, \frac{b_{|M|}}{m_{|M|}} \right\} \quad (12)$$

where

$$b_k = \frac{SI(m_k)}{\max_{m \in M} SI(m)} \quad (13)$$

In many contexts, users rate an item only once. In this case, the value of Rating per Item is only 0 or 1, and consequently even $b_k = \{0, 1\}$. Although $ItemAttr_{u_i}(m)$ will have only two values, 0 or 1, it will be very useful to represent the interest of the user for the recommendation technique. It provides a very quick way to compare the movies rated by two users, as defined in the *UserSignature* and *UserKindredness* described below.

Following Yager and Reformat's definitions 6 and 7, the fuzzy relation *UserSignature* US_{u_i} is constructed as follows:

$$US_{u_i}(s, m) = RatPop_{u_i}(s) \times ItemAttr_{u_i}(m) \quad (14)$$

which, for a single item m_k and rating s_j , is calculated as:

$$US_{u_i}(s_j, m_k) = \min\{RatPop_{u_i}(s_j), ItemAttr_{u_i}(m_k)\} \quad (15)$$

TABLE 2. Fuzzy relation as a signature of the user u_k .

	m_1	m_2	m_3	m_4	m_5	m_6	m_7	m_8	m_9	m_{10}	Rating Popularity
1	1/3										1/3
2			2/3						2/3		2/3
3						1/3					1/3
4					2/3					2/3	2/3
5				3/3			3/3	3/3			3/3
Item Attractiveness	1	0	1	1	1	1	1	1	1	1	

The second and final step in computing the Fuzzy User Signature consists in transforming each cell marked with an X in Mat_{u_i} in a value calculated with Eq. 15. Since in our context a user can only assign one or no rating to an item, $ItemAttr_{u_i}$ of any item can only be 0 or 1, so each UserSignature value is effectively obtained by applying Eq. 11. In our example, starting from the matrix in Table 1, the transformation results in the matrix in Table 2.

Nevertheless, in environments in which users may interact multiple times with the same items (such as social tagging systems), the FUS model not only captures rating behaviors, but also explicitly identifies items of higher engagement. In these contexts, $ItemAttr_{u_i}$ would take non-binary values, ensuring that two users are not considered similar if they exhibit different degrees of interest in their shared items. This results in a more nuanced user model.

Once the user signatures are calculated for each user, they are compared using the *similarity* metric, inspired by the concept of *User Kindredness* [11]. Given two sets A and B representing two different users u_i and u_j respectively, the kindredness measure $K(A, B)$ is formally defined as follows:

$$K(A, B) = \frac{|A \cap B|}{|A|} \quad (16)$$

where $|\cdot|$ denotes the cardinality of a set. It should be noted that the kindredness measure is asymmetric, meaning that $K(A, B)$ refers to the point of view of user u_i , described by the set A . This is natural, since in reality we make comparisons with other people based on what we know about ourselves. Moreover, this asymmetry addresses the imbalance in information density across users. When a novice user with few ratings is compared to an experienced user, the lower cardinality of the user signature of the former ensures that kindredness from the novice's perspective is high, while it remains low from the expert's point of view, as a sparse profile cannot fully represent the well-defined tastes of a power user. This promotes systemic fairness in the recommendations and leads to more personalized and reliable results. Since in our user model A and B are fuzzy sets, the intersection is computed using a *T-norm*, i.e., a triangular norm that models fuzzy intersection. Hence, the User Kindredness UK of user u_i with respect to user u_j is defined as:

$$UK(u_i, u_j) = \frac{|T_{norm}(US_{u_i}(s, m), US_{u_j}(s, m))|}{|US_{u_i}(s, m)|} \quad (17)$$

Our proposed method employs the element-wise *min* operator as the *T-norm* and *scalar cardinality* as a measure

of the cardinality of a finite fuzzy set, defined as the sum of its membership degrees. The choice of the *min* operator penalizes discrepancies between the signatures and ensures that the calculated kindredness reflects behavioral alignment. We denote the user kindredness measure of users u_i and u_j as the *similarity* between the two users $sim(u_i, u_j)$, defined as follows:

$$sim(u_i, u_j) = \frac{sum(T_{norm}(US_{u_i}(s, m), US_{u_j}(s, m)))}{sum(US_{u_i}(s, m))} \quad (18)$$

where *sum* indicates the sum of all the elements in the matrix.

Cardinality of fuzzy sets can be determined for a specific value of α -cut. In the context of scalar cardinality, given α , the cardinality is calculated by summing only the membership values that are greater than α . In our application domain, a higher value of α decreases both the value of the numerator (the sum of the elements of the T-norm, which represents a measure of the common interests of the two users), and the denominator (the sum of the elements of his/her user signature, which represents a measure of his/her interests). In other words, a high value of α can be interpreted as a stricter requirement on the minimum level of interest necessary for considering two resources as similar. In section V, the effect of different values of α will be considered. Once the similarity measure is calculated for each ordered pair of users, a neighborhood $N_{u_i}^k$ for the active user u_i can be established, including the k users with highest similarity with respect to u_i .

B. PREDICTED SCORE AND TOP-X ITEMS

In order to extract the top-x items that will be recommended to the active user u_i , the FUS method identifies the predicted rating that he/she would assign to each item $m \in M$. Then, the x items with highest predicted rating are recommended to the user. These ratings are obtained by employing the well-known Resnick prediction formula, adapted to our method:

$$p_{u_i}^m = \bar{r}_{u_i} + \frac{\sum_{u_j \in N_{u_i}^k} (r_{u_j}^m - \bar{r}_{u_j}) \cdot sim(u_i, u_j)}{\sum_{u_j \in N_{u_i}^k} sim(u_i, u_j)} \quad (19)$$

Here, $p_{u_i}^m$ is the predicted preference for item m by user u_i , $N_{u_i}^k$ is the top-k neighborhood of user u_i , $\bar{r}_{u_i}$ and $\bar{r}_{u_j}$ are the average of all the ratings given by users u_i and u_j respectively, and $sim(u_i, u_j)$ is the similarity between the users. It is important to note that in our method, if a user u_j has not voted for item m , the whole term of the summation is set to zero, as substituting zero in the term $r_{u_j}^m$ would distort the estimate.

V. EVALUATION

The proposed recommendation approach FUS was tested in a movie recommendation system, consisting of users, movies as items and ratings in the range (1 - 5), a scale widely adopted by major platforms such as Amazon, TripAdvisor and Google Play. The FUS approach is compared with different reference solutions: the probabilistic solution (PF) [12], the

fuzzy-genetic approach (GIM) proposed by Al-Shamri and Bharadwaj [13] and the collaborative filtering method (CF) [14]. The PF approach is similar to the proposed FUS method, but it uses the concept of probability instead of fuzziness. The GIM method was selected as it proposes a similar approach to predict user interest; the CF is used as an experimental baseline as it is one of the most common recommendation approach.

A. DATASET

For the validation of our recommendation system, we chose the *Movielens 100k* dataset [32], released in April 1998 by *GroupLens*,¹ due to its relevance in the field of movie recommendation systems. It comprises 100,000 ratings from 943 users across 1,682 movies. It also includes additional information about users (such as age, gender, occupation, zip code), and movies (such as genre, year of publication, IMDb URL). The preferences are memorized as tuples (*user,item,rating,timestamp*).

The dataset was filtered in order to maintain consistency and address data imbalance, by removing the movies with fewer than 20 votes and selecting the top 497 users based on the number of votes (as in [12]). The resulting dataset's composition is detailed in Table 3.

TABLE 3. Characteristics of the experimental dataset.

# of users	497
# of items	903
# of topics	18
# of ratings	79432

B. PERFORMANCE METRICS

The metrics used for the evaluation are the following:

- *Mean Absolute Error (MAE)*: Measures the average absolute deviation of predicted ratings from actual ratings.
 - *MAE_{data}* considers all ratings in the validating set, providing an overall system performance measure:

$$MAE_{data} = \frac{1}{N} \sum_{(i,j) \in T} |\hat{r}_{ij} - r_{ij}| \quad (20)$$

Here, N is the total number of ratings in the validation set, (i, j) represents the user-item pair, $\hat{r}_{ij}$ is the predicted rating, r_{ij} is the actual rating and T is the validating set of users.

- *MAE_i* calculates the average absolute deviation for each user u_i , focusing on items they rated:

$$MAE_i = \frac{1}{|R_i|} \sum_{r_{ij} \in R_i} |\hat{r}_{ij} - r_{ij}| \quad (21)$$

In this formula, $|R_i|$ represents the number of ratings made by user u_i , and the summation is over the ratings given by that user.

¹<http://grouplens.org>

- MAE_{users} represents the overall user experience by averaging MAE_i over all users:

$$MAE_{users} = \frac{1}{N_u} \sum_{i=1}^{N_u} MAE_i \quad (22)$$

N_u is the number of users in the validation set, and MAE_i is the MAE calculated for each user u_i . The main difference with the MAE_{data} metric that it assigns equal weight to all users, instead of being dominated by very active users, which have rated more items. This means that MAE_{users} penalizes a recommender system that performs poorly on a subgroup of users.

- **Root Mean Square Error (RMSE):** Measures the square root of the average squared differences between predicted and actual ratings:

$$RMSE = \sqrt{\frac{\sum_{(i,j) \in T} (\hat{r}_{ij} - r_{ij})^2}{N}} \quad (23)$$

Again, (i, j) represents the user-item pair in the validation set T , $\hat{r}_{ij}$ is the predicted rating, r_{ij} is the actual rating, and N is the total number of ratings. The difference between MAE and RMSE is that the latter emphasizes larger errors, since they are squared before being averaged.

- **Coverage Rate (CR):** Measures the capability of the system to suggest new items to users. It is defined as follows:

$$CR = \frac{|\hat{Y}|}{|V|} \quad (24)$$

i.e. the ratio of $|\hat{Y}|$, the number of predictions made in the validation phase, to $|V|$, the number of resource-user pairs in the validation set. It is important to note that a high CR does not indicate high prediction accuracy. Rather, it reflects the capability of the system to cover a wide range of items when producing recommendations.

C. EXPERIMENTS AND RESULTS

The experiments have been designed to evaluate the proposed FUS method individually to characterize its behavior with respect to the parameters k and α , and then to compare it with the reference solutions — PF, CF, GIM.

We employ *10-fold cross-validation* on the dataset: it is first divided in 10 folds, 9 for training and 1 for testing, and for each validation a different test fold is selected. For each metric, we take the average across all the folds. This process is replicated for different neighbor values k to characterize the behavior of the FUS method with respect to k , and for different α value, to find the best configuration of the FUS method.

The experimental evaluation is structured into three sections. The first subsection displays the performance stability of our system by presenting the variation of the Mean Absolute Error (MAE) across different validation folds,

for different neighborhood sizes. The second subsection will aggregate the data to examine how each performance metric responds to changes in the neighborhood size and α value, identifying the optimal configuration. Lastly, the third subsection will compare the FUS solution against PF, GIM and CF to assess any enhancements achieved by implementing the User Signature method.

1) AVERAGE MAE FOR DIFFERENT NEIGHBORHOOD SIZES

Fig. 2 illustrates the mean and standard deviation of the MAE for different neighborhood sizes, as obtained through the 10-fold cross-validation process, with the MAE on the y-axis and the number of neighbors on the x-axis. For each number of neighbors, a bar representing the average and an error bar indicating the standard deviation are shown. We can observe that the standard deviation remains constant across the different numbers of neighbors, denoting that the recommender's performance remains stable regardless of the neighborhood configuration. Moreover, its low value suggests that the MAE is consistent across different folds of data. We can also observe that the average MAE tends to decrease as the number of neighbors increases, indicating that the accuracy of the predictions improves when more information is included in the user model.

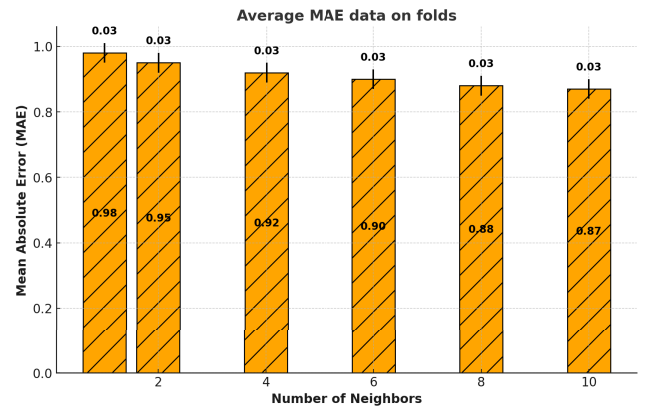


FIGURE 2. Average MAE with standard deviation error bars for different number of neighbors in the recommendation system.

2) PERFORMANCE OF THE FUS METHOD

The performance of the FUS method was evaluated with respect to different number of neighbors $k = \{1, 2, 4, 6, 8, 10, \dots, 48, 50\}$ and different values of $\alpha = \{0, 0.7, 0.8, 0.9\}$. Only high α values have been chosen, since in the case of the Movielens dataset, the user signature values are distributed mainly around 0.8, so lower alpha values do not influence the performance. From the charts in Fig. 3 we can see the different behaviors. Specifically, we observe a similar trend both for the MAE_{data} , computed using 20 and for the MAE_{users} , computed using 22.

The reason for using both metrics is to gain a comprehensive understanding of the system's effectiveness. whereas MAE_{data} gives us a global picture of the accuracy, MAE_{users} ensures that this accuracy is consistent across different users.

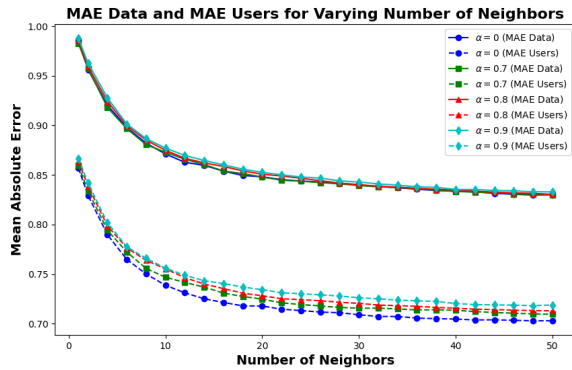


FIGURE 3. The influence of neighborhood number and alpha value on the MAE.

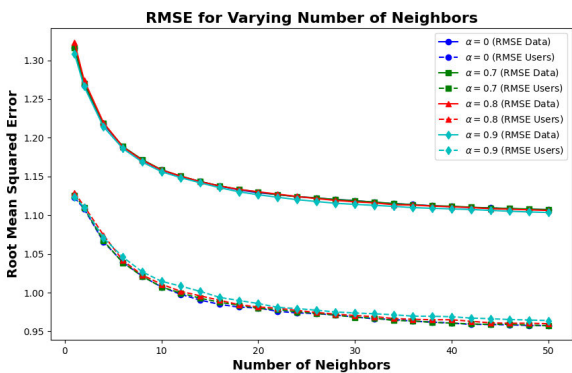


FIGURE 4. The influence of neighborhood number and α value on the RMSE.

With $\alpha = 0$, the MAE_{data} , starting from the highest value 0.985, decreases as it increases the value of *neighborhood* up to a value of 0.83 when the number of neighbors is 50. Also the MAE_{users} shows a descending trend as it starts with the highest value 0.857 and decreases up to a value of 0.703.

This behavior can be explained by the fact that a higher number of neighbors allows the system to utilize more information, thus increasing the accuracy of the predictions.

whereas the MAE_{data} remains consistent across different α values, the MAE_{users} increases along with α , and the difference is more noticeable when the number of neighbors is greater than 10. This can be interpreted as a consequence of the loss of information obtained by removing membership values from the similarity calculation.

The RMSE, as depicted in Fig. 4, behaves similarly to the MAE as it decreases as the number of neighbors reaches 50. The difference is in the value of the error, which is slightly higher: in fact, the highest value is 1.316 and the lowest value is 1.107. This can be explained by the fact that our method usually predicts positive ratings, so when the true rating is positive (≥ 3) the error is very low, but when it is negative (≤ 2) the error becomes quite large. These large errors are emphasized by the RMSE metric. A similar behavior is shown by $RMSE_{users}$, only with lower values overall, starting from 1.123 and reaching 0.957. In this case, the difference

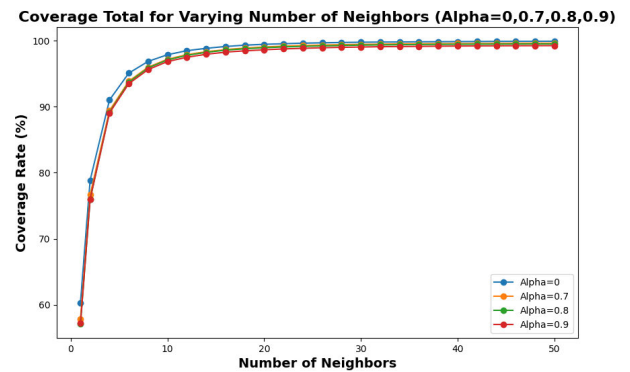


FIGURE 5. The influence of neighborhood number on the coverage rate.

between different α values is much less significant, meaning that the magnitude of the errors remains similar as α varies.

As shown in Fig. 5, the Coverage Rate (CR) grows with the number of neighbors—the higher the better. Specifically, it starts at slightly lower than 60% for 1 neighbor and increases rapidly up to 14 neighbors, then stabilizing after 30 neighbors, and finally reaching around 99% at 50 neighbors. These values remain consistent as α varies, demonstrating the stability of the algorithm. As discussed for the MAE and RMSE, the fact that the system provides recommendations for a larger portion of the dataset when the number of neighbors is higher can be attributed to the inclusion of more data points in the calculation of the predictions. In other words, more neighbors means including the preferences of more similar users, leading to a richer user model and generally to a more robust recommendation system.

From these observations, we can conclude that the growth trend of CR—fast up to 10-15 neighbors, then gradually slowing down until fully stabilizing after 30—highlights the importance of selecting the appropriate number of neighbors. However, going over a certain threshold is unnecessary, as performance remains almost constant beyond that point.

3) COMPARISON

The performance of the FUS approach is compared with the PF approach [12], based on probability, the GIM approach [13], and collaborative filtering CF, in Fig. 6 - 7 [14]. Since the reference solutions perform worse with more than 10 neighbors, we compare the four methods up to this number of neighbors. Furthermore, both the PF and CF approaches consider this number of neighbors in the reference papers [12], [13]. However, it is important to note that for higher numbers of neighbors, the performance of the FUS approach improves. As indicated by Fig. 6, the FUS method presents lower MAE values compared to the other solutions when the number of neighbors is 4 and above. Moreover, while the MAE values of the other methods tend to plateau or worsen beyond two neighbors, the FUS method steadily improves. These lower error margins between predicted and true ratings demonstrate a superior

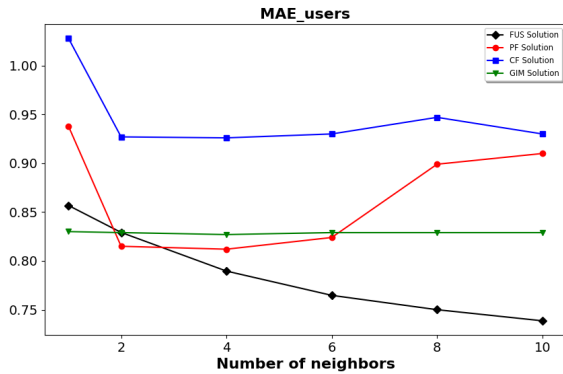


FIGURE 6. The effect of the neighborhood number on MAE values comparison.

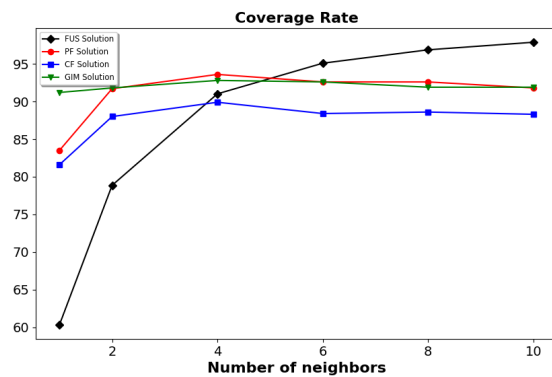


FIGURE 7. The effect of the neighborhood number on CR values comparison.

accuracy of FUS compared to the reference methods, as it is able to better leverage the increasing amount of information and consistently improves with a larger number of neighbors.

Regarding the Coverage Rate, the FUS method shows better performance than the other solutions. As depicted in Fig. 7, our methods surpasses the others when the number of neighbors is 6 and above. Our solution reaches a value of 97.9%, whereas the PF and GIM solutions reach a maximum value of 91.8%. Similarly to the MAE results, the Coverage Rate of the other methods tends to plateau, whereas FUS continues to improve as the number of neighbors increases. However, this means that FUS may need more information to achieve optimal results, and thus selecting the optimal value of the parameter k is crucial.

D. DISCUSSION

The results obtained from the proposed Fuzzy User Signature-based solution (FUS) demonstrate good performances. Specifically, both the MAE and RMSE exhibit very low values, indicating high precision in recommendations, whereas the coverage rate achieves high values, suggesting a broad applicability of the system to various user preferences.

However, it's important to note that the high coverage rate, exceeding 99%, primarily reflects the system's ability to match user preferences, which may not directly

correlate with the actual accuracy or appropriateness of each recommendation. Therefore, these results, while promising, must be interpreted considering the inherent limitations of recommendation system metrics.

Furthermore, our method shows improved performance compared to both the PF [12] and GIM [13] approach, particularly in terms of MAE and Coverage Rate. This suggests a more precise recommendation capability with fewer errors thanks to the use of the fuzziness to estimate similarity among users. In addition, the experimental analysis highlights that FUS benefits more than the other methods from the inclusion of additional neighbors: while competing approaches tend to plateau or even deteriorate, FUS consistently improves as k increases, reaching a limit for very high values of k . Nevertheless, choosing the parameter k is a balancing act, as a larger number of neighbors can enhance predictive accuracy but at the cost of higher computational time for generating recommendations and greater memory requirements to store neighbor sets for each user.

Key distinctions between our method and the probabilistic and fuzzy-genetic approach include:

- Our method employs a *Fuzzy User Signature* model and *User Kindredness* measure for constructing user-model matrices and computing user similarities, in contrast to the probability-based models of the PF approach and the hybrid feature models of the GIM approach.
- whereas probabilistic and fuzzy-genetic solutions often focus on broader categories or additional information about the items, our fuzzy approach is centered on rating assigned to individual items, providing a more detailed basis for recommendations.

Finally, two key limitations of the FUS method can be identified:

- Scalability challenges, since the batch processing phase requires updating user signatures and similarities as the number of users and items grows. This can be mitigated by exploiting sparsity, clustering, and candidate-neighbor selection, which we plan to explore in future work.
- As a form of collaborative filtering, our method also suffers from the cold start problem. This can be mitigated with complementary approaches [33], and as soon as a small fuzzy signature becomes available, it can be compared with target fuzzy signatures derived from user clusters to provide a starting point for the recommendation process.

VI. CONCLUSION

This work introduces a new recommendation approach based on the fuzzy signature method, namely FUS method, applied to movie recommendation. We adapted the concepts of user signature and kindredness of a tagging system to a movie recommendation system in order to obtain a synthetic indicator of the user preferences. This enables an efficient calculation of user similarity, from which it is possible to

establish a neighborhood and compute the predicted ratings. Importantly, the method is generalizable and can be applied to any context where users assign tags or ratings to resources, such as adaptive e-learning systems, online marketplaces, and streaming services.

A comparison between the FUS method and probabilistic approach, collaborative filtering approach and fuzzy genetic approach, indicates a significant improvement. Key innovations include a novel user-activity representation and streamlined activity matrix construction.

Future work will focus on exploring parallel and distributed computation frameworks for the offline processing phase (e.g., GPU and cluster computing) and on the application of this approach to other fields, including Internet of Things and digital health [34], [35], cultural heritage [36], [37], security [38] and e-learning, specifically for the recommendation of learning materials to define adaptive courses. Finally, our method can be improved in handling negative ratings to enhance user experience and complementary approaches can be explored to handle cold start scenarios.

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